

Harsh Bordekar

AI Scientist - Physics Models

Physics-Informed ML for Simulation, Surrogate Modelling & Uncertainty Quantification
PhD Candidate, DLR & TU Braunschweig — Aerospace Composites & Structural Simulation

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🌐 [Harsh's LinkedIn](#)

📍 Braunschweig, Germany

🔗 github.com/harshbordekar-stack

🏠 Permanent Resident - Germany

PROFILE

Simulation and machine-learning research engineer with 7+ years developing physics-based models for aerospace structures, including 3+ years applying ML and surrogate modelling directly to simulation problems. PhD work builds a probabilistic, physics-based multiscale surrogate framework for CFRP composites, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure from ply to structural scale and replacing computationally expensive full-scale FEM with fast, physics-consistent predictions. The framework is evaluated against a certification-aligned failure criterion derived as part of the same work. Runs large-scale simulation campaigns on HPC/SLURM using domain-specific solvers (ABAQUS, Nastran/Patran) to generate the high-fidelity data these models depend on, and applies Monte Carlo uncertainty quantification to characterise prediction confidence ahead of design decisions. This is direct, hands-on experience with the UQ/out-of-distribution estimation this role treats as a first-class concern. Also built a physics-based CNN surrogate for on-demand computational homogenization, exposed through a tool-calling interface, replacing full-scale FEM homogenization runs with fast inference. Comfortable across the applied-ML stack (PyTorch, TensorFlow, scikit-learn) and in Python/Linux/HPC environments day to day, with a track record of verifying and technically reviewing externally produced simulation models against defined acceptance criteria, and of explaining simulation and ML concepts to both engineering and non-technical audiences.

Core Competencies: ML for Simulation & Surrogate Modelling, Physics-Informed Machine Learning, Multiscale / Multi-Fidelity Damage Modelling, Uncertainty Quantification & Out-of-Distribution Estimation, High-Fidelity Simulation Data Generation, FE Solvers (ABAQUS, Nastran/Patran), HPC-Scale Simulation Campaigns (SLURM), Model Verification & Validation, Physics-Based CNN / Surrogate Modelling, Explainable AI, Python

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

Physics-informed ML for simulation & surrogate modelling (PhD)

Feb 2021 - Present

- (PhD) Developed a probabilistic, physics-based multiscale surrogate framework for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales, replacing [x] full-scale FEM runs with a physics-consistent surrogate and estimating probability of failure directly from the simulation-generated dataset.

Tools: Python, ABAQUS, FEM, HPC/SLURM, Multiscale Modelling

- (PhD) Derived a certification-aligned leakage failure criterion mapping combined strain/load states to leakage onset, bridging progressive-damage models and system-level tank simulations and providing the physics-based evaluation standard against which the surrogate framework's coverage, accuracy, and failure modes are assessed.

Tools: ABAQUS, Python, FEM post-processing using C++

- Performed Monte Carlo uncertainty quantification over the multiscale simulation outputs to generate leakage-probability-vs-strain relationships, characterising prediction confidence and enabling risk-informed design decisions for next-generation CFRP pressure vessels. This is direct UQ/out-of-distribution experience applied to physics-simulation outputs.

Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS

- Ran large-scale thermal and structural simulation campaigns on HPC infrastructure to evaluate thermal-protection effectiveness for the Callisto reusable-rocket programme's metal upper housing bay, using domain-specific solvers to generate high-fidelity simulation data under production-scale compute constraints.

Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran

- Built a physics-based CNN surrogate for microstructure-informed computational homogenization, wrapped in an agentic, tool-calling interface that accepts natural-language queries and microstructure images, retrieves context via a hierarchical RAG pipeline over 50 materials documents, and returns homogenized/effective material properties, replacing full-scale FEM homogenization runs (hours) with on-demand inference (seconds). Retrieval/prediction faithfulness validated at 93% against ground-truth homogenization results on a held-out set of more than 100 cases.

Tools: Python, Physics-Based CNN, Hierarchical RAG, LLM Agent Framework, Image Processing

- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes arising from data gaps or modelling limitations and issuing documented sign-off-style findings.

Tools: ABAQUS, Nastran, Python

- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM). This demonstrated the ability to communicate simulation and ML concepts to both technical and non-technical audiences.
Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance

Explainable AI & physics-based fatigue/defect-tolerance assessment

DLR-Braunschweig

Nov 2019 - Jan 2021

- Developed a CT-based, explainable-AI defect-detection framework achieving **91%** classification accuracy (**89** precision / **91** recall) on **1200** CT scans of additively manufactured titanium parts, cutting manual review time to **<0.5 sec per image**, with SHAP, Grad-CAM and LIME outputs used to support engineering sign-off. (*Published: Journal of Intelligent Manufacturing, 2025.*)
Tools: Python (scikit-learn, TensorFlow), SHAP, Grad-CAM, LIME, CT image processing

- Assessed fatigue and defect tolerance of aerospace-grade titanium components using small-defect fatigue models (Murakami, El-Haddad), coupling defect measurements with numerical simulation to translate physics-based fracture-mechanics criteria into allowable-defect acceptance limits.

Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy), FEM (ABAQUS)

Student Research Assistant

Institute of Static and Dynamic

Leibniz University Hannover

Sep 2018 - Oct 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates from image data, generating spatially resolved defect metrics used directly as high-fidelity inputs for probabilistic structural simulation models assessing manufacturing-induced variability.

Tools: MATLAB, Python (scikit-learn, TensorFlow), image processing, NumPy/SciPy

SKILLS SUMMARY

ML for Simulation & Surrogate Modelling	Physics-Informed Machine Learning Multiscale / Multi-Fidelity Damage Modelling Physics-Based CNN Surrogates Computational Homogenization
Uncertainty & Evaluation	Uncertainty Quantification (Monte Carlo) Verification & Validation Physics-Based Evaluation Against Certification Criteria Failure-Mode Diagnosis
Simulation Solvers & HPC	ABAQUS Nastran/Patran Linear & Non-linear FEM HPC / SLURM (Large-Scale Simulation Campaigns) Analytical Stress Methods (Niu, Bruhn)
ML & AI	PyTorch TensorFlow Scikit-learn Explainable AI (SHAP, Grad-CAM, LIME) Hugging Face Transformers (Fine-tuning)
LLMs & Agentic Systems	Retrieval-Augmented Generation (RAG) LangChain LangGraph Model Context Protocol (MCP) Tool-Calling
Programming Languages	Python C++ FORTRAN MATLAB
Data Analysis & Scientific Computing	NumPy Pandas Matplotlib Seaborn Plotly SALib SciPy
APIs & Backend Development	RESTful APIs FastAPI JSON
Version Control & Reproducibility	Git Docker Linux
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

 2nd Prize: Best Paper Award	DLR (German Aerospace Center) 2026
 3rd Prize: Best Paper Award	DLR (German Aerospace Center) 2024
PhD Candidate, Computational Methods in Engineering	DLR & TU Braunschweig, Germany Feb 2021 – 2026 (expected)
Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures	
M.Sc. Computational Methods in Engineering	Leibniz University Hannover, Germany April 2018 – Dec 2020
Focus: Numerical Methods, Scientific Machine Learning, Deep Learning	

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [DOI](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [DOI](#)

REFERENCES

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg